

SOFTWARE REQUIREMENTS SPECIFICATION

Complaint Management Portal

Professional Project Documentation

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Document Type Software Requirements Specification

Project	Complaint Management Portal
Version	1.0
Status	Baseline / Academic Project

1. Introduction

1.1 Purpose

This Software Requirements Specification establishes the functional and non-functional requirements for a **Complaint Management Portal**. The system provides a centralized web-based platform through which users can register complaints, track their status, communicate with the concerned department, and receive resolution updates.

The document provides a clear baseline for the design, implementation, testing, and evaluation of the Complaint Management Portal.

1.2 Problem Statement

In many organizations and institutions, complaints are submitted through manual processes such as written applications, emails, phone calls, or direct communication. These approaches can make complaint tracking difficult, cause delays in resolution, and provide limited visibility into complaint status.

The proposed Complaint Management Portal digitizes the complaint-handling process by providing complaint registration, categorization, assignment, tracking, status updates, notifications, feedback, and administrative management through a centralized system.

1.3 Scope

The Complaint Management Portal provides the following major capabilities:

- User registration and authentication.
- Complaint submission and registration.
- Complaint categorization and prioritization.
- Complaint tracking and status monitoring.
- Assignment of complaints to responsible departments or staff.
- Complaint status updates.
- User notifications regarding complaint progress.

- Communication between users and complaint-handling staff.
- Complaint history management.
- Feedback and complaint-resolution rating.
- Administrative management of users, departments, and complaints.
- Reporting and complaint analytics.

1.4 Intended Audience

This SRS is intended for:

- Students and project developers.
 - Software testers.
 - Faculty/project guides.
 - System administrators.
 - Complaint-handling staff.
 - Department heads.
 - End users/citizens/students/employees.
 - Other project stakeholders.
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2. Overall Description

2.1 Product Perspective

The Complaint Management Portal is a web-based application consisting of a presentation layer, backend/API layer, database, notification service, and administrative management module.

The system provides a centralized platform where complaints can be submitted and managed throughout their complete lifecycle, from registration to final resolution.

2.2 Major Functions

The major functions of the system are:

- User registration and authentication.
- User profile management.
- Complaint registration.
- Complaint categorization.
- Complaint priority selection.
- Complaint assignment to departments/staff.
- Complaint status tracking.
- Complaint search and filtering.

- Complaint history.
- Notifications and updates.
- User-staff communication.
- Complaint resolution.
- Feedback and rating.
- Administrative management.
- Complaint reports and analytics.

2.3 User Classes

User / Complainant

The user can:

- Create an account.
- Login securely.
- Submit complaints.
- Upload supporting information where applicable.
- View complaint details.
- Track complaint status.
- Receive notifications.
- Communicate regarding complaints.
- Provide feedback after resolution.

Complaint Handling Staff

The staff can:

- Login securely.
- View assigned complaints.
- Update complaint status.
- Add remarks or resolution details.
- Communicate with complainants.
- Mark complaints as resolved.

Administrator

The administrator can:

- Manage users.
- Manage departments.

- Manage staff accounts.
- View and manage complaints.
- Assign complaints.
- Monitor complaint status.
- Generate reports.
- Manage system data.

2.4 Assumptions

- Users have access to a modern web browser and network connection.
- Users provide valid information while registering and submitting complaints.
- Complaint-handling staff are available to process assigned complaints.
- The database is available and maintained properly.
- Notifications depend on the availability of the configured notification service.
- Complaint resolution time depends on the responsible department or staff.

3. Functional Requirements

ID	Requirement	Specification
FR-01	Registration	Allow new users to create an account.
FR-02	Authentication	Authenticate registered users and provide secure logout.
FR-03	Profile Management	Allow users to view and update their profile information.
FR-04	Complaint Registration	Allow authenticated users to submit new complaints.
FR-05	Complaint Category	Allow users to select an appropriate complaint category.
FR-06	Complaint Priority	Allow complaints to be assigned an appropriate priority level.
FR-07	Complaint Details	Store complaint title, description, category, priority, date, and user information.
FR-08	Complaint Assignment	Allow administrators to assign complaints to appropriate departments or staff.

ID	Requirement	Specification
FR-09	Status Tracking	Allow users to view the current status of their complaints.
FR-10	Status Update	Allow authorized staff to update complaint status.
FR-11	Complaint Search	Support searching and filtering complaints using relevant criteria.
FR-12	Notifications	Notify users about important complaint status changes.
FR-13	Communication	Allow authorized users and staff to exchange complaint-related messages or remarks.
FR-14	Resolution	Allow staff to provide resolution details and mark complaints as resolved.
FR-15	Feedback	Allow users to provide feedback or rating after complaint resolution.
FR-16	Complaint History	Maintain the history of complaint status changes and actions.
FR-17	Administration	Allow administrators to manage users, staff, departments, and complaints.
FR-18	Reports	Provide complaint statistics and administrative reports.
FR-19	Error Handling	Validate inputs and provide meaningful error messages.

4. Non-Functional Requirements

ID	Attribute	Specification
NFR-01	Performance	Provide responsive complaint submission, search, tracking, and management under expected project load.
NFR-02	Security	Use secure password hashing, authentication, authorization, and role-based access control.
NFR-03	Usability	Provide a simple, responsive, consistent, and easy-to-navigate interface.

ID	Attribute	Specification
NFR-04	Reliability	Handle invalid input, database failures, and common service failures gracefully.
NFR-05	Scalability	Support growth in users, complaints, departments, and system records.
NFR-06	Maintainability	Keep frontend, backend, database, notification, and administrative modules modular.
NFR-07	Privacy	Restrict complaint and user information to authorized users.
NFR-08	Availability	The system should be available whenever users need to submit or track complaints, subject to infrastructure availability.
NFR-09	Auditability	Maintain records of important complaint actions and status changes.

5. External Interface Requirements

5.1 User Interface

The interface should include:

- Login and registration page.
- User dashboard.
- Complaint submission form.
- Complaint list.
- Complaint details page.
- Complaint tracking/status page.
- Notification section.
- Profile settings.
- Feedback form.
- Staff dashboard.
- Administrator dashboard.
- Complaint reports and analytics.

5.2 Software Interfaces

The system may use the following technologies:

- **Frontend:** React.js or equivalent modern web framework.

- **Backend:** Node.js with Express.js or equivalent REST API framework.
- **Database:** MongoDB, PostgreSQL, or equivalent database.
- **Authentication:** JWT/session-based authentication.
- **Notification Service:** Email or web-based notification service.
- **Optional Storage:** Cloud/local storage for supporting complaint documents.

5.3 Hardware Interfaces

No specialized hardware is required.

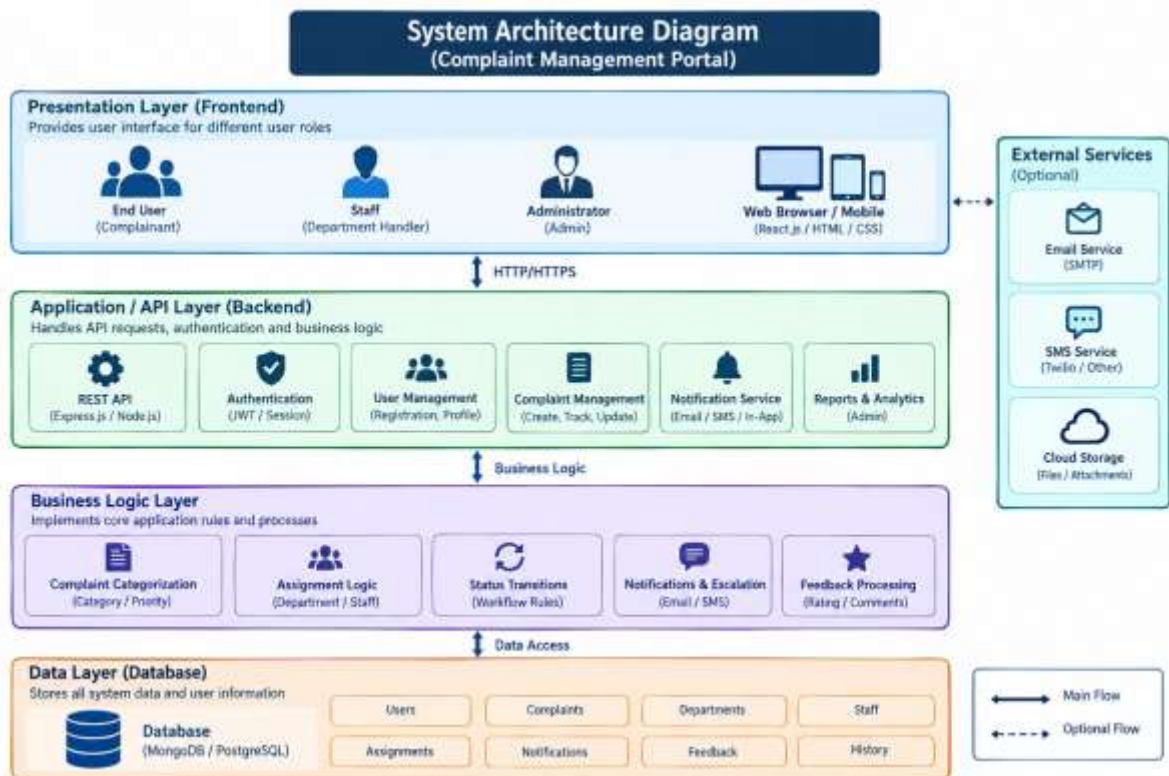
The system can operate on standard:

- Desktop computers.
- Laptops.
- Tablets.
- Smartphones.

6. Data Requirements

Entity	Representative Attributes
User	user_id, name, email, password_hash, phone, role, created_at
Complaint	complaint_id, user_id, title, description, category, priority, status, created_at
Department	department_id, department_name, description
Staff	staff_id, name, email, department_id, role, created_at
Assignment	assignment_id, complaint_id, staff_id, assigned_at
Status History	history_id, complaint_id, old_status, new_status, remarks, timestamp
Notification	notification_id, user_id, complaint_id, message, read_status, created_at
Feedback	feedback_id, complaint_id, user_id, rating, comment, created_at
Message	message_id, complaint_id, sender_id, message, timestamp

7. System Architecture



The Complaint Management Portal consists of the following major layers:

7.1 Presentation Layer

Provides the responsive web interface for users, staff, and administrators.

7.2 Application/API Layer

Handles:

- Authentication.
- Complaint registration.
- Complaint management.
- Assignment.
- Status updates.
- Notifications.
- Feedback.
- Reports.
- Administration.

7.3 Business Logic Layer

Handles complaint-processing rules such as:

- Complaint categorization.
- Priority handling.
- Assignment.
- Status transitions.
- Resolution workflow.
- Feedback processing.

7.4 Data Layer

Stores:

- User information.
- Complaint records.
- Staff information.
- Department information.
- Complaint history.
- Notifications.
- Feedback.
- Messages.

7.5 Notification Layer

Handles notifications related to:

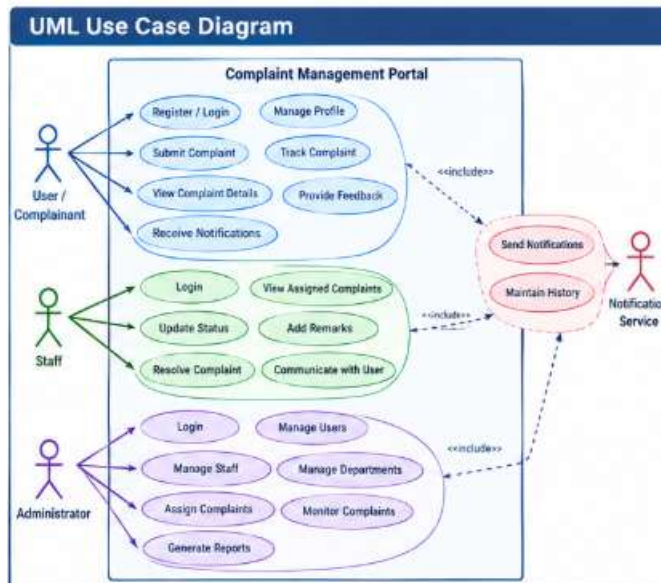
- New complaint registration.
- Complaint assignment.
- Status changes.
- Resolution.
- Other important complaint updates.

8. Use Case Specifications

ID	Use Case	Actor	Primary Flow	Outcome
UC-01	Register / Login	User	Create account or authenticate.	Authenticated session.
UC-02	Manage Profile	User	Update personal information.	Profile saved.

ID	Use Case	Actor	Primary Flow	Outcome
UC-03	Submit Complaint	User	Enter complaint details and submit.	Complaint registered.
UC-04	Track Complaint	User	Open complaint and check status.	Current complaint status displayed.
UC-05	View Complaint Details	User/Staff/Admin	Open a complaint record.	Complaint information displayed.
UC-06	Assign Complaint	Administrator	Assign complaint to department/staff.	Complaint assigned.
UC-07	Update Status	Staff	Change complaint status and add remarks.	Complaint status updated.
UC-08	Resolve Complaint	Staff	Add resolution details and close complaint.	Complaint marked resolved.
UC-09	Send Notification	System	Generate notification after important actions.	User receives update.
UC-10	Provide Feedback	User	Submit rating/comment after resolution.	Feedback stored.
UC-11	View Complaint History	User/Staff/Admin	Open complaint history.	Previous actions displayed.
UC-12	Manage Users	Administrator	Create, update, deactivate users/staff.	User records managed.
UC-13	Manage Departments	Administrator	Add/update/delete departments.	Department records managed.
UC-14	Generate Reports	Administrator	Select report criteria.	Complaint statistics/report generated.

9. UML Use Case Diagram



The UML Use Case Diagram represents the interaction between the major actors and the Complaint Management Portal.

Primary Actors

- User / Complainant
- Complaint Handling Staff
- Administrator
- Notification Service

Major Use Cases

User:

- Register/Login
- Manage Profile
- Submit Complaint
- Track Complaint
- View Complaint Details
- Receive Notifications
- Provide Feedback

Staff:

- Login
- View Assigned Complaints
- Update Status
- Add Remarks

- Resolve Complaint
- Communicate with User

Administrator:

- Login
- Manage Users
- Manage Staff
- Manage Departments
- Assign Complaints
- Monitor Complaints
- Generate Reports

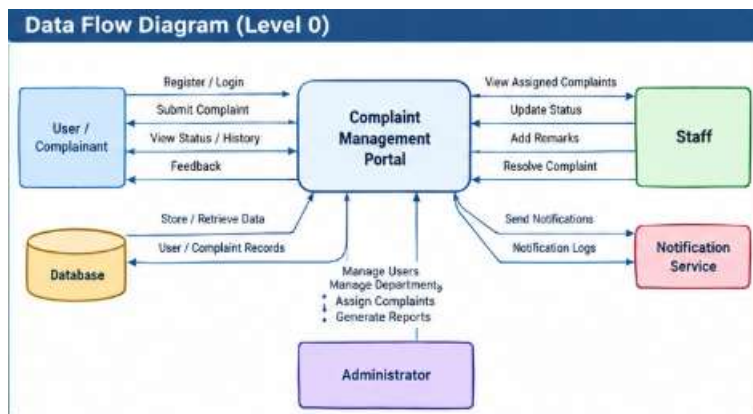
System:

- Send Notifications
 - Maintain Complaint History
 - Validate Complaint Data
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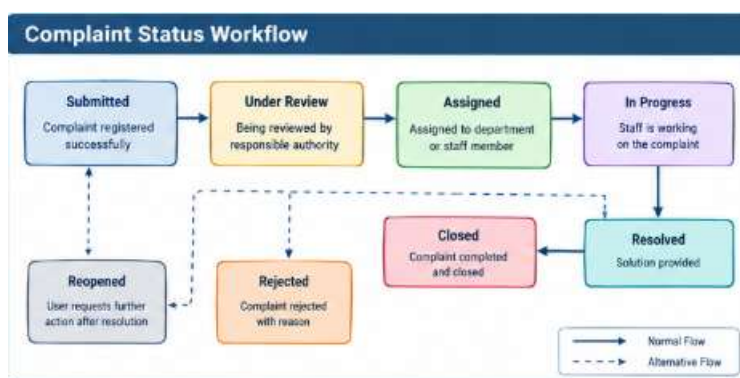
10. Security Requirements

- Passwords shall never be stored in plaintext.
- Protected operations shall require authentication.
- Role-based authorization shall be implemented for users, staff, and administrators.
- Users shall only be allowed to access their authorized complaint information.
- Staff shall only manage complaints assigned to them or permitted by their role.
- Administrator functions shall be protected through role-based access control.
- User input shall be validated and sanitized at API boundaries.
- Sensitive information shall be protected during transmission.
- HTTPS shall be used in production.
- Important complaint actions and status changes should be logged for auditing.
- Unauthorized access attempts should be rejected appropriately.

Data flow Diagram :



11. Complaint Status Workflow



A complaint may progress through the following statuses:

Submitted → Under Review → Assigned → In Progress → Resolved → Closed

A complaint may also be marked as:

Rejected / Reopened

Status Description

Status	Description
Submitted	Complaint has been successfully registered.
Under Review	Complaint is being reviewed by the responsible authority.
Assigned	Complaint has been assigned to a department or staff member.
In Progress	Staff is actively working on the complaint.
Resolved	A solution has been provided.
Closed	Complaint has been completed and closed.
Reopened	User has requested further action after resolution.

Status	Description
Rejected	Complaint has been rejected with an appropriate reason.

12. Future Enhancements

The following enhancements may be added in future versions:

- AI-based complaint categorization.
 - Automatic complaint priority prediction.
 - AI-based complaint routing to departments.
 - Chatbot for complaint registration and status queries.
 - Sentiment analysis of complaint descriptions.
 - Automatic duplicate complaint detection.
 - Predictive complaint-resolution time.
 - Mobile application.
 - SMS and email notifications.
 - Advanced complaint analytics.
 - Geographic complaint visualization.
 - Automated escalation for delayed complaints.
 - Multilingual complaint submission.
 - Voice-based complaint registration.
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13. Acceptance Criteria

The Complaint Management Portal shall be considered acceptable when:

- Users can register and authenticate successfully.
- Authenticated users can submit complaints.
- Complaint information is stored correctly.
- Users can view and track their complaints.
- Administrators can assign complaints to appropriate staff/departments.
- Staff can view assigned complaints.
- Authorized staff can update complaint status.
- Staff can provide resolution details.
- Users receive appropriate complaint updates.

- Users can provide feedback after resolution.
 - Complaint history is maintained.
 - Administrators can manage users, departments, and complaints.
 - Reports can be generated from complaint records.
 - Unauthorized users cannot access protected information.
 - Invalid input and common system failures are handled gracefully.
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14. Conclusion

The **Complaint Management Portal** provides a structured, centralized, and transparent platform for managing complaints digitally. The system reduces dependence on manual complaint-handling processes and provides users with better visibility into complaint status and resolution.

The proposed system supports complaint registration, categorization, assignment, tracking, communication, resolution, notifications, feedback, and administrative reporting. This SRS establishes a professional baseline for system design, implementation, testing, evaluation, and future enhancement.

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